

PROJECT PROPOSAL

Nub-Journal
A Web-Based Blog Publishing and Management System

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1. Introduction

Nub-Journal is a lightweight, web-based blog publishing platform proposed as a course project for the Database Management System (DBMS) module. The system allows registered users to securely log in, create and manage their own blog posts, and publish them for public readers to browse and read. The project is intentionally scoped as a small, focused application so that the relational database design, normalization, and query logic behind it can be implemented and demonstrated thoroughly within the timeframe of a course project.

2. Problem Statement

Individuals and small communities who want to share articles, notes, or personal write-ups often rely on heavyweight third-party platforms that offer little control over data structure, ownership, or customization. There is a need for a minimal, self-contained publishing system where authentication, content management, and content delivery are handled by a well-structured relational database, giving the owner full control over their data while keeping the application simple to deploy and maintain.

3. Objectives

- Design a normalized relational database schema capable of supporting user accounts and blog content.
- Implement secure user authentication (registration, login, and session/token-based access control).
- Provide full Create, Read, Update, and Delete (CRUD) operations for blog posts, restricted to their owners.
- Allow authors to publish or unpublish posts, and allow public visitors to read published posts.
- Demonstrate sound database practices: primary/foreign key constraints, indexing, and efficient queries.

4. Scope and Modules

The system is divided into three core modules. The scope is deliberately kept compact to allow each module to be implemented and tested with proper database backing.

4.1 Authentication Module

Handles user registration and login, password hashing and verification, and session/token management (e.g., JWT or server-side sessions). Ensures that only authenticated users can access content-management features, while public blog reading remains open to all visitors.

4.2 Blog CRUD Module

Allows an authenticated author to create new blog posts, edit or update existing ones, view a list of their own posts, and delete posts they own. All operations are scoped to the logged-in user via foreign-key ownership checks at the database level.

4.3 Blog Publish & Read Module

Lets an author change a post's status between draft and published. Published posts become visible on a public listing page and an individual post-detail page, where any visitor (authenticated or not) can read the content.

5. System Design Overview

5.1 Database Design

The schema is normalized to Third Normal Form (3NF) to avoid redundancy and maintain data integrity. Two core entities drive all three modules, related by a one-to-many relationship: one user can author many blog posts.

Table	Key Fields	Purpose
Users	user_id (PK), name, email (unique), password_hash, created_at	Stores account credentials used by the Authentication module.
Blogs	blog_id (PK), user_id (FK → Users), title, slug, content, status, created_at, updated_at, published_at	Stores post content and status; used by CRUD and Publish/Read modules.

Relationship: Users (1) ■■ owns ■■> (M) Blogs, enforced via a foreign key with cascading rules so that ownership and status checks happen at the database layer, not just in application code.

5.2 Proposed Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, JavaScript (Bootstrap for layout/styling)
Backend	Node.js with Express.js (equivalent frameworks such as PHP or Django are acceptable substitutes)
Database	MySQL (relational, supports constraints, indexing, and joins required for this module)
Auth	bcrypt for password hashing, JWT for stateless session tokens

6. Expected Outcome

A fully functional, deployable blog publishing system demonstrating: a normalized relational schema, secure authentication, owner-scoped CRUD operations, and a public read interface — together showing practical command of relational database design and SQL query implementation in a real application context.

7. Project Timeline

Phase	Activities	Duration
1	Requirement analysis & ER/schema design	Week 1
2	Database implementation & Authentication module	Week 2
3	Blog CRUD module development	Week 3
4	Blog Publish & Read module + frontend integration	Week 4
5	Testing, debugging & documentation	Week 5

8. Conclusion

Nub-Journal offers a compact but database-driven scope ideal for demonstrating core DBMS concepts — schema normalization, referential integrity, and CRUD query design — within a real, working web application. The team is confident that the proposed three-module structure can be completed and thoroughly tested within the available timeframe.

